

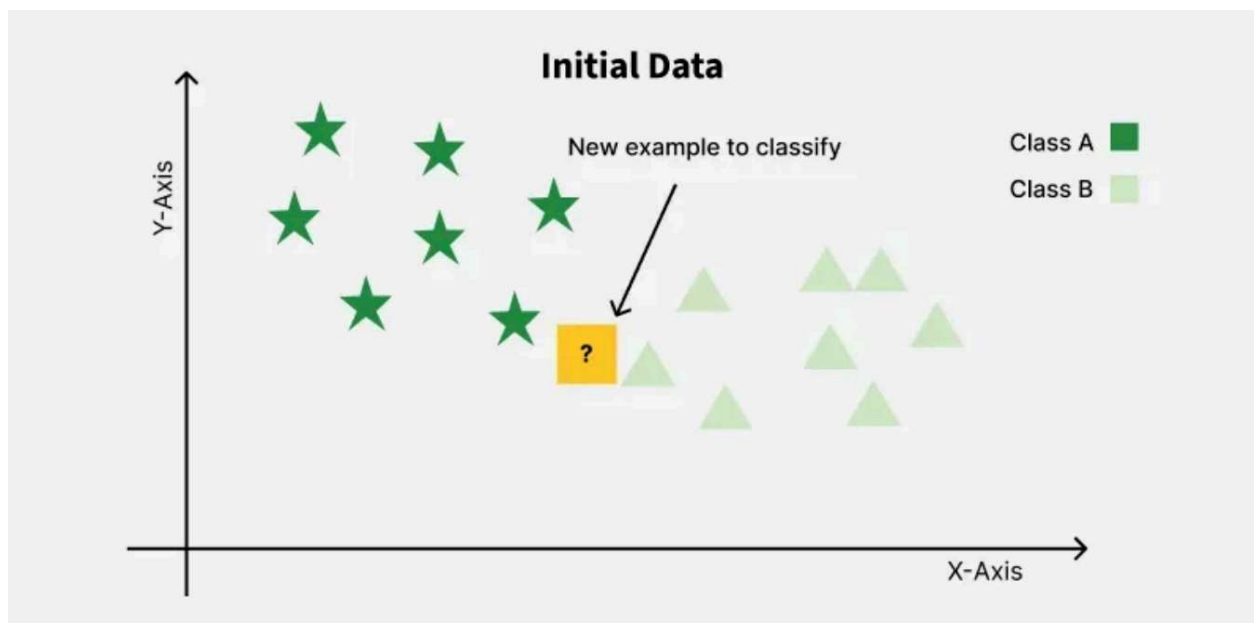
K-Nearest Neighbours · Naive Bayes · Feature Scaling

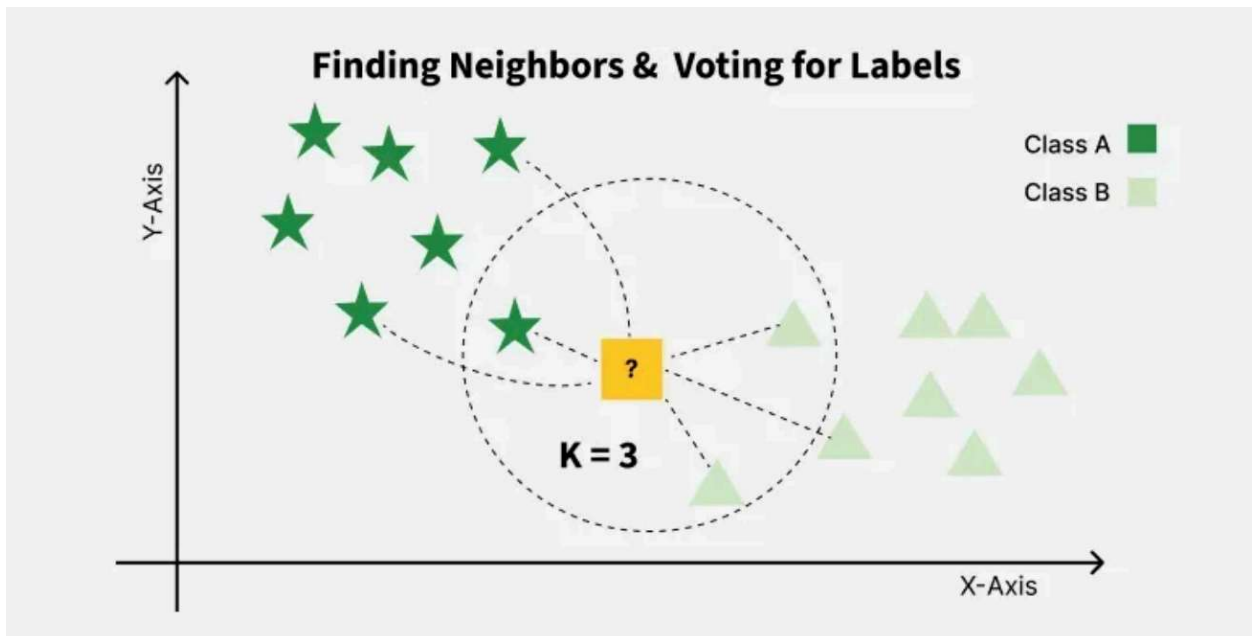
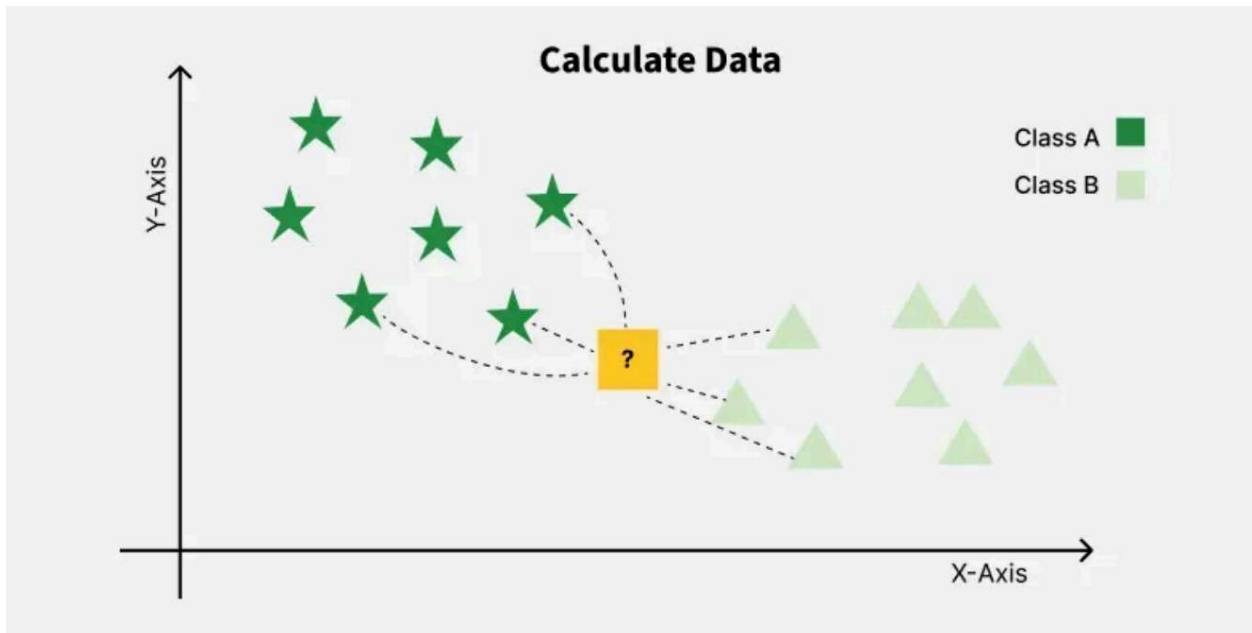
K-Nearest Neighbours (KNN)

KNN is a lazy learning algorithm — it does no explicit training. Instead, it memorises the entire training dataset and defers all computation to prediction time. When a new data point arrives, KNN searches through all stored examples, finds the K closest ones, and assigns the majority class label.

The word lazy has a formal meaning here: there is no generalisation step, no learned model parameters, no weight updates. This is the opposite of eager learners like Logistic Regression or Neural Networks, which train a model once and discard the training data. KNN keeps everything.

K-Nearest Neighbors (KNN) is a supervised machine learning algorithm generally used for classification but can also be used for regression tasks. It works by finding the "k" closest data points (neighbors) to a given input and makes a predictions based on the majority class (for classification) or the average value (for regression). Since KNN makes no assumptions about the underlying data distribution it makes it a non-parametric and instance-based learning method.





K-Nearest Neighbors is also called as a lazy learner algorithm because it does not learn from the training set immediately instead it stores the entire dataset and performs computations only at the time of classification.

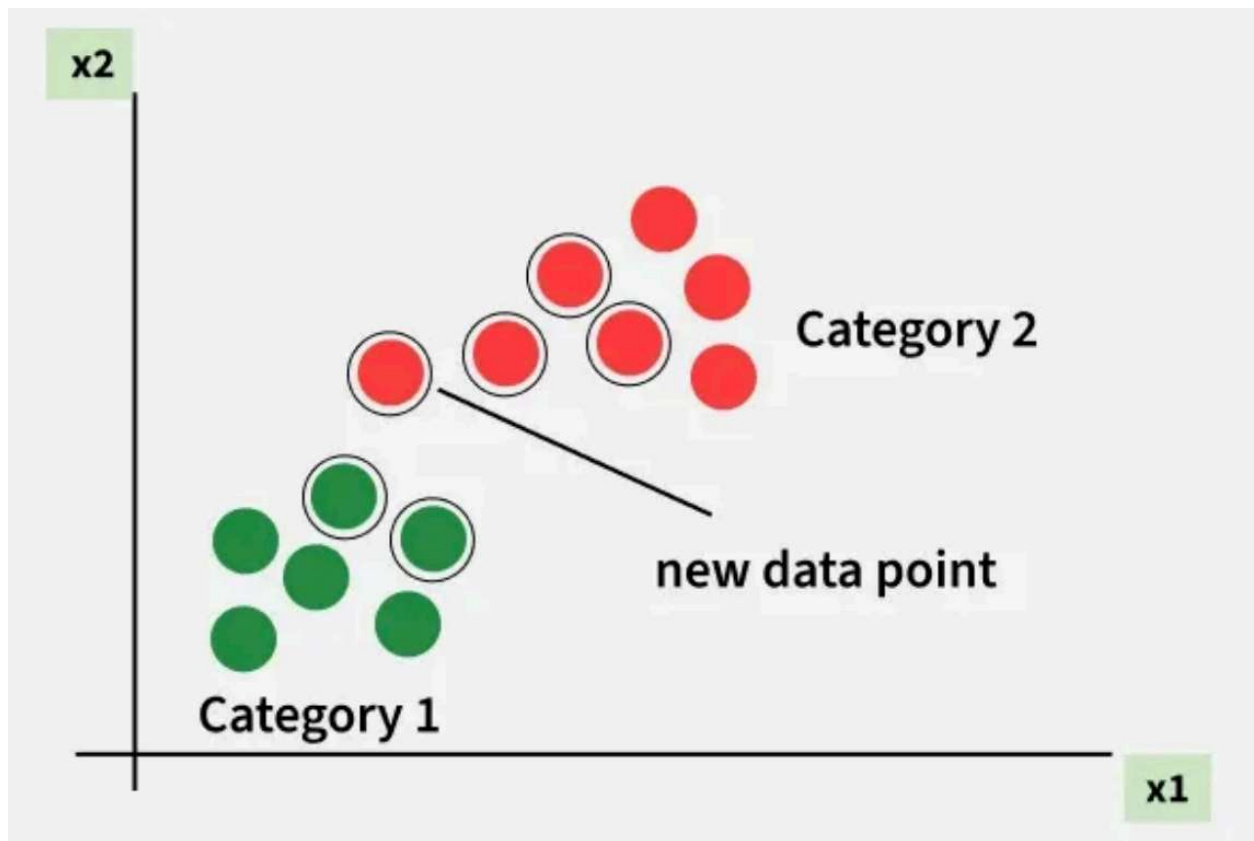
For example, consider the following table of data points containing two features:

The new point is classified as Category 2 because most of its closest neighbors are red circles. KNN assigns the category based on the majority of nearby points. The image shows how KNN predicts the category of a new data point based on its closest neighbours.

The blue circles represent Category 1 and the red circles represent Category 2.

The new data point checks its closest neighbors (circled points).

Since the majority of its closest neighbors are red circles (Category 2) KNN predicts the new data point belongs to Category 2.



How KNN Works — Step by Step

Step 1 — Compute distances. For each test point, calculate its distance to every training point using a chosen metric (Euclidean, Manhattan, etc.).

Step 2 — Sort and select. Sort all training points by distance (ascending) and pick the top K.

Step 3 — Vote. For classification, take a majority vote among the K neighbours. For regression, take the mean of their values.

Step 4 — Assign. The new point receives the winning class label.

What is 'K' in K Nearest Neighbour?

In the k-Nearest Neighbours algorithm k is just a number that tells the algorithm how many nearby points or neighbors to look at when it makes a decision.

Example: Imagine you're deciding which fruit it is based on its shape and size. You compare it to fruits you already know.

If $k = 3$, the algorithm looks at the 3 closest fruits to the new one.

If 2 of those 3 fruits are apples and 1 is a banana, the algorithm says the new fruit is an apple because most of its neighbors are apples.

How to choose the value of k for KNN Algorithm?

The value of k in KNN decides how many neighbors the algorithm looks at when making a prediction.

Choosing the right k is important for good results.

If the data has lots of noise or outliers, using a larger k can make the predictions more stable. But if k is too large the model may become too simple and miss important patterns and this is called underfitting.

So k should be picked carefully based on the data.

Statistical Methods for Selecting k

Cross-Validation: Cross-Validation is a good way to find the best value of k is by using k-fold cross-validation. This means dividing the dataset into k parts. The model is trained on some of these parts and tested on the remaining ones. This process is repeated for each part. The k value that gives the highest average accuracy during these tests is usually the best one to use.

Process

Split data into multiple folds.

Train on some folds.

Test on remaining fold.

Repeat several times.

Compute average accuracy.

Select k with highest average accuracy.

Cross-validation tests multiple values of k and chooses the one that performs best.

k	Average Accuracy
1	88%
3	91%
5	94%
7	93%
9	92%

```

from sklearn.neighbors import KNeighborsClassifier
from sklearn.model_selection import cross_val_score

for k in range(1, 21):
    model = KNeighborsClassifier(n_neighbors=k)
    scores = cross_val_score(model, X, y, cv=5)

    print(k, scores.mean())

```

Elbow Method: In Elbow Method we draw a graph showing the error rate or accuracy for different k values. As k increases the error usually drops at first. But after a certain point error stops decreasing quickly. The point where the curve changes direction and looks like an "elbow" is usually the best choice for k .

Odd Values for k : It's a good idea to use an odd number for k especially in classification problems. This helps avoid ties when deciding which class is the most common among the neighbors.

Distance Metrics

This is formatted as code

KNN uses distance metrics to identify nearest neighbor, these neighbors are used for classification and regression task. To identify nearest neighbor we use below distance metrics:

Distance Metric	Formula	Best Used When
Euclidean Distance	Straight-line distance	Continuous numerical data
Manhattan Distance	Distance along grid paths	Data with outliers or grid-like movement
Minkowski Distance	Generalized form of Euclidean & Manhattan	Flexible choice
Hamming Distance	Count of mismatched positions	Binary, categorical, or text data

Metric	Formula	Use When
Euclidean	$\sqrt{\sum (x_i - y_i)^2}$	Default choice — continuous numeric features
Manhattan	$\sum x_i - y_i $	Grid data, more robust to outliers than Euclidean
Minkowski	$(\sum x_i - y_i ^p)^{1/p}$	Generalises both; $p=2$ is Euclidean, $p=1$ is Manhattan
Hamming	count of differing positions	Categorical / binary features, strings

Which Distance Metric Should You Use?

Data Type	Recommended Metric
Numeric continuous features	Euclidean

Data Type	Recommended Metric
Numeric with outliers	Manhattan
Want flexibility	Minkowski
Binary/Categorical data	Hamming

Working of KNN algorithm

The K-Nearest Neighbors (KNN) algorithm operates on the principle of similarity where it predicts the label or value of a new data point by considering the labels or values of its K nearest neighbors in the training dataset.

Step 1: Selecting the optimal value of K

K represents the number of nearest neighbors that needs to be considered while making prediction.

Step 2: Calculating distance

To measure the similarity between target and training data points Euclidean distance is widely used. Distance is calculated between data points in the dataset and target point.

Step 3: Finding Nearest Neighbors

The k data points with the smallest distances to the target point are nearest neighbors.

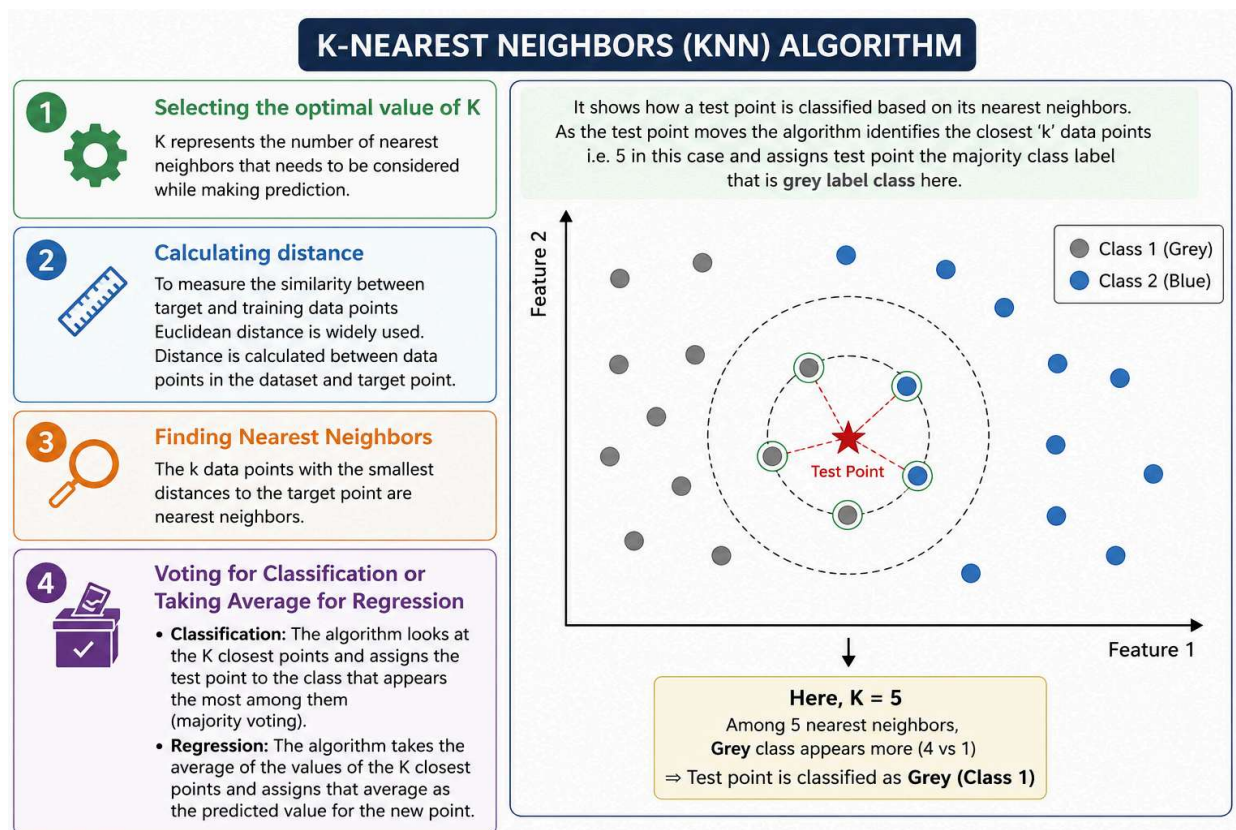
Step 4: Voting for Classification or Taking Average for Regression

When you want to classify a data point into a category like spam or not spam, the KNN algorithm looks at the K closest points in the dataset.

These closest points are called neighbors. The algorithm then looks at which category the neighbors belong to and picks the one that appears the most. This is called majority voting.

In regression, the algorithm still looks for the K closest points. But instead of voting for a class in classification, it takes the average of the values of those K neighbors. This average is the predicted value for the new point for the algorithm.

It shows how a test point is classified based on its nearest neighbors. As the test point moves the algorithm identifies the closest 'k' data points i.e. 5 in this case and assigns test point the majority class label that is grey label class here.



Choosing the Right K

K is the single most important hyperparameter in KNN. There is no universally correct value — it depends on your dataset. Here is what happens at different extremes:

K	Behaviour	Technical Term
K = 1	Boundary exactly wraps every training point	Maximum overfitting / high variance
K small (2–5)	Complex, jagged decision boundary	Overfitting — noisy predictions
K moderate	Smoother boundary — generalises well	Balanced bias-variance tradeoff
K large	Very smooth, nearly linear boundary	Underfitting — high bias
K = N	Always predicts the majority class	Maximum underfitting

Bias-Variance Tradeoff in KNN

Bias = how wrong your model is on average (underfitting). Variance = how much your model changes with different training sets (overfitting).

In KNN: small K → low bias, high variance (follows training data too closely). Large K → high bias, low variance (too generalised). The sweet spot is the K that minimises total error on unseen data — found via cross-validation.

Strengths and Weaknesses

Strengths	Weaknesses
No training time — <code>fit()</code> just stores data	Slow prediction: $O(N*d)$ per query
Naturally handles multi-class problems	Memory-heavy: stores all training data
Non-parametric: no assumptions about data shape	Irrelevant features hurt accuracy ('curse of dimensionality')
Works well with small, clean datasets	Requires feature scaling — sensitive to magnitude

KNN Implementation

```
import numpy as np
from collections import Counter # a a a a b b. c c c c c  a:4 b:2 c:5
# ■ Manual KNN — see exactly what sklearn does ■
class KNNClassifier:
    def __init__(self, k=3):#cons
        self.k = k
    def fit(self, X, y):
        self.X_train = np.array(X) # just store everything
        self.y_train = np.array(y)
    def _euclidean(self, a, b):
        return np.sqrt(np.sum((a - b) ** 2))
    def predict_one(self, x):
        dists = []
        for xt in self.X_train:
            dists.append(
                self._euclidean(x, xt)
            )
        k_idx = np.argsort(dists)[:self.k] # K closest

        k_labels = [self.y_train[i] for i in k_idx]

        return Counter(k_labels).most_common(1)[0][0] # majority vote
    def predict(self, X):
        return [self.predict_one(x) for x in X]
```

Complete work flow

`fit()`

↓

Store training data

↓

`predict()`

↓

For each test point

↓

Calculate distances

↓

Sort distances

↓

Pick K nearest neighbors

↓

Collect labels

↓

Majority vote

↓

Return prediction

```
# ■ sklearn KNN with cross-validation to find best K ■■■
from sklearn.neighbors import KNeighborsClassifier
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split, cross_val_score
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import accuracy_score, classification_report

#load data set
iris = load_iris()
X, y = iris.data, iris.target

#split for train and test
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, ran

#create a scalar
scaler = StandardScaler()

#scale training data
X_train = scaler.fit_transform(X_train) # fit on train only
```

```
#scale test data
X_test = scaler.transform(X_test) # apply same transform to test

# Find best K
k_range = range(1, 33)

# cv for eachk
cv_scores = [
    cross_val_score(KNeighborsClassifier(n_neighbors=k), X_train, y_train, c
    for k in k_range
]
#find best k
best_k = list(k_range)[cv_scores.index(max(cv_scores))]
print(f'Best K={best_k} CV Acc={max(cv_scores):.4f}')

#create final knn model
knn = KNeighborsClassifier(n_neighbors=best_k)
knn.fit(X_train, y_train)
```

Best K=3 CV Acc=0.9500

▼ **KNeighborsClassifier** ⓘ ?
 KNeighborsClassifier(n_neighbors=3)

```
import matplotlib.pyplot as plt

plt.plot(k_range, cv_scores, marker='o')
plt.xlabel("K")
plt.ylabel("CV Accuracy")
plt.show()
```

[Show hidden output](#)

Naive Bayes — Probability Classifier

The Probability Foundation

Unlike KNN which uses geometry, Naive Bayes uses probability theory. It is a generative classifier — it models the probability distribution of each class and uses that to make predictions. This makes it extremely fast to train (a single pass through data) and very efficient at prediction time.

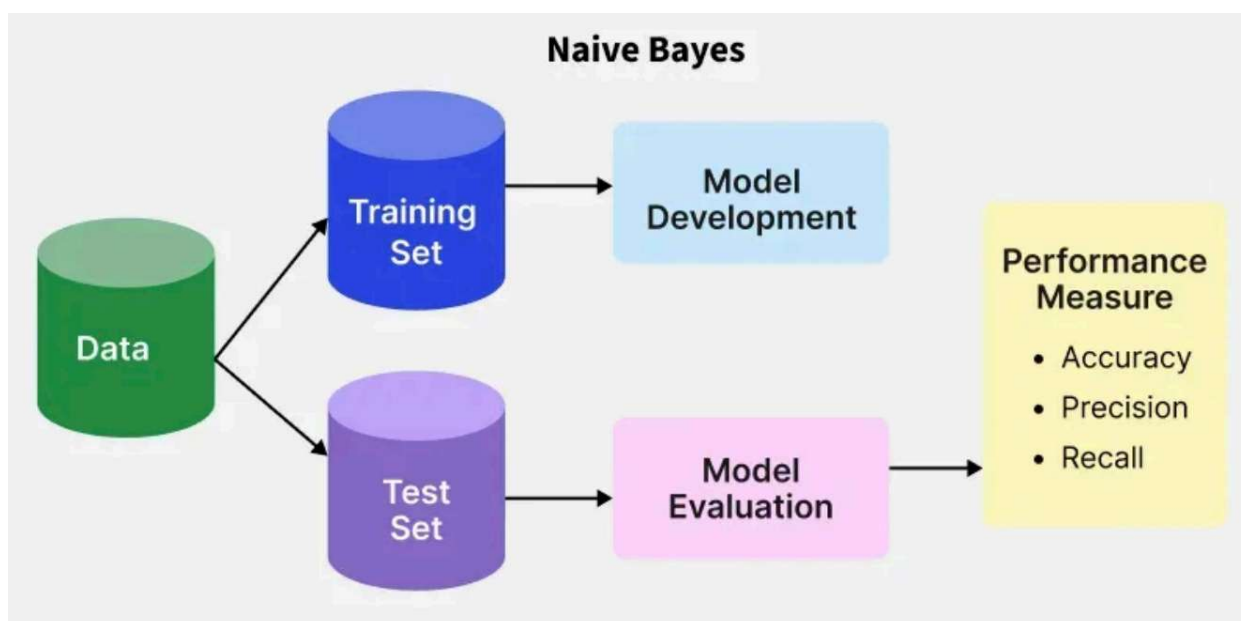
Bayes Theorem

Bayes Theorem tells us how to update a prior belief when we observe new evidence:

$$P(\text{Class} \mid \text{Features}) = [P(\text{Features} \mid \text{Class}) \times P(\text{Class})] / P(\text{Features})$$

Term	Name	Spam Example
$P(\text{Class} \mid \text{Features})$	Posterior — what we want	Prob this email is spam, given its words
$P(\text{Features} \mid \text{Class})$	Likelihood	If it IS spam, how likely are these words?
$P(\text{Class})$	Prior belief	40% of all emails in training set were spam
$P(\text{Features})$	Evidence (normaliser)	Same for all classes — we can ignore it

Naive Bayes is a probabilistic machine learning algorithms based on the Bayes Theorem. It is popular method for classification applications such as spam filtering and text classification.



Why Naive? The Independence Assumption

Naive Bayes computes $P(\text{Features} \mid \text{Class})$ by assuming all features are conditionally independent given the class label. This means:

$$P(\text{word1}, \text{word2}, \text{word3} \mid \text{spam}) = P(\text{word1} \mid \text{spam}) \times P(\text{word2} \mid \text{spam}) \times P(\text{word3} \mid \text{spam})$$

In reality, words like 'free' and 'prize' co-occur in spam emails — they are correlated, not independent. The algorithm ignores this correlation. That is the naive assumption. Despite being technically wrong, it works remarkably well in text classification because we care about the correct ranking of classes, not the exact probability values.

Three Flavours of Naive Bayes

Variant	Assumption	Best For
GaussianNB	Features follow a normal (Gaussian) distribution	Continuous numeric: sensor data, Iris dataset
MultinomialNB	Features are discrete counts or frequencies	Text: word counts, TF-IDF document classification
BernoulliNB	Features are binary (word present = 1 or absent = 0)	Short text, spam detection, document retrieval

Naive Bayes Code — Spam Classifier with Pipeline

"Win ₹10000 now"

"Claim your free prize"

"Are you coming to class?"

"Let's meet tomorrow"

Machine Learning cannot understand text directly.

Text

↓

Numbers

↓

Machine Learning Model

↓

Spam / Ham

```
from sklearn.naive_bayes import MultinomialNB, GaussianNB
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.pipeline import Pipeline
from sklearn.model_selection import train_test_split
from sklearn.metrics import classification_report
```

```
# ■ Sample email dataset
emails = [
    'Win a free iPhone now click here', # spam
    'Congratulations you won a lottery prize', # spam
    'FREE cash prize claim within 24 hours', # spam
    'Limited offer buy now get 50 percent off', # spam
```

```
'Meeting at 3pm tomorrow please confirm', # ham
'Quarterly report attached for review', # ham
'Can you send me the project files thanks', # ham
'Team lunch tomorrow noon at cafeteria', # ham
'Urgent your account has been suspended', # spam
'Hi please review the attached document', # ham
]
```

```
labels = [1,1,1,1,0,0,0,0,1,0] # 1=spam 0=ham
```

```
X_train,X_test,y_train,y_test = train_test_split(
emails, labels, test_size=0.3, random_state=42)
```

```
# ■ Pipeline: TF-IDF vectorise -> classify ■
# TF-IDF: rare informative words get higher weight than common ones
#aas we mentioned we need to convert text to numbers for that we are going t
#IDF = Inverse Document Frequency
# alpha=1.0 is Laplace smoothing — prevents zero-probability words
```

```
pipe = Pipeline([
('tfidf', TfidfVectorizer(lowercase=True, stop_words='english')),
('nb', MultinomialNB(alpha=1.0)),
])

pipe.fit(X_train, y_train)
y_pred = pipe.predict(X_test)
print(classification_report(y_test, y_pred, target_names=['Ham', 'Spam']))
```

	precision	recall	f1-score	support
Ham	0.50	1.00	0.67	1
Spam	1.00	0.50	0.67	2
accuracy			0.67	3
macro avg	0.75	0.75	0.67	3
weighted avg	0.83	0.67	0.67	3

Text Message

↓

TF-IDF

↓

Numerical Features

↓

Naive Bayes

↓

Spam/Ham

This pipeline first converts text into numerical TF-IDF features. TF-IDF gives higher importance to informative rare words and lower importance to common words.

The transformed features are then passed to a Multinomial Naive Bayes classifier, which calculates probabilities for each class and predicts whether a message is Ham or Spam.

Laplace smoothing ($\alpha=1.0$) prevents zero probabilities for unseen words. Finally, `classification_report` evaluates model performance using precision, recall, and F1-score.

```
# ■ Inspect learned probabilities
nb_model = pipe.named_steps['nb']
vectorizer = pipe.named_steps['tfidf']
feature_names = vectorizer.get_feature_names_out()
import numpy as np

MultinomialNB()
TfidfVectorizer(stop_words='english')
['24' '3pm' '50' 'attached' 'buy' 'cafeteria' 'cash' 'claim' 'click'
 'confirm' 'document' 'files' 'free' 'hi' 'hours' 'iphone' 'limited'
 'lunch' 'meeting' 'noon' 'offer' 'percent' 'prize' 'project' 'review'
 'send' 'team' 'thanks' 'tomorrow' 'win']
```

```
# Top spam words (highest log-probability in spam class)
spam_log_probs = nb_model.feature_log_prob_[1] # class index 1 = spam
```

```
top_spam_idx = np.argsort(spam_log_probs)[-10:][::-1]
print('Top spam words:', [feature_names[i] for i in top_spam_idx])
```

```
[-3.25194799 -3.60208154 -3.23244149 -3.60208154 -3.23244149 -3.60208154
 -3.25194799 -3.25194799 -3.1829458  -3.60208154 -3.60208154 -3.60208154
 -3.02535532 -3.60208154 -3.25194799 -3.1829458  -3.23244149 -3.60208154
 -3.60208154 -3.60208154 -3.23244149 -3.23244149 -3.25194799 -3.60208154
 -3.60208154 -3.60208154 -3.60208154 -3.60208154 -3.60208154 -3.1829458 ]
Top spam words: ['free', 'win', 'iphone', 'click', 'percent', 'offer', 'buy',
```

```
# Predict new emails with confidence
new = [' your its a new job its been great well working this is examp
for email, pred, prob in zip(new, pipe.predict(new), pipe.predict_proba(new))
print(f'["SPAM" if pred==1 else "HAM"] conf={prob[pred]*100:.1f}% {email}')
```

```
[HAM] conf=57.1% your its a new job its been great well working this
```

Feature Scaling — Why It Matters

The Problem Without Scaling

Suppose your dataset has two features: Age (range 20–65) and Salary (range 20,000–1,20,000). KNN computes Euclidean distance between two people as:

$$d = \sqrt{(age1 - age2)^2 + (salary1 - salary2)^2}$$

An age difference of 5 years contributes 25 to the sum. A salary difference of just 500 rupees contributes 2,50,000. The salary term completely dominates. KNN becomes a salary-only classifier and ignores age entirely. Feature scaling fixes this

StandardScaler — Standardisation

Centres the data around zero mean with unit standard deviation. Each feature independently becomes: how many standard deviations away from the mean is this value?

$$z = (x - \text{mean}(x)) / \text{std}(x)$$

Result: feature mean = 0, feature std = 1. Values are typically in the range [-3, +3]. Outliers are present but scaled down. This is the most commonly used scaler.

MinMaxScaler — Normalisation

Squeezes all values into a fixed range, typically [0, 1]. Preserves the original distribution shape but is sensitive to outliers (a single extreme value compresses all other values into a tiny range).

Use when: the algorithm explicitly needs $[0,1]$ inputs, or you know there are no outliers in the data. Common in neural networks with sigmoid activation functions.

Uses the median and interquartile range (IQR) instead of mean and std. Because the median and IQR are not affected by extreme values, this scaler is robust when your data has significant outliers.

Comparison of Scalers

The Golden Rule: Fit on Train, Transform Both

The scaler must learn its parameters (mean, std, min, max) from the training data only. You then apply those same learned parameters to transform the test set.

If you fit the scaler on the full dataset (train + test), the test mean and std influence the transformation. Your model has effectively peeked at test data statistics before evaluation. This is called data leakage and inflates your reported accuracy.

Correct pattern: `scaler.fit transform(X train)` then `scaler.transform(X test)`

```
import numpy as np, pandas as pd
from sklearn.preprocessing import StandardScaler, MinMaxScaler, RobustScaler
from sklearn.model_selection import train_test_split
import matplotlib.pyplot as plt
```

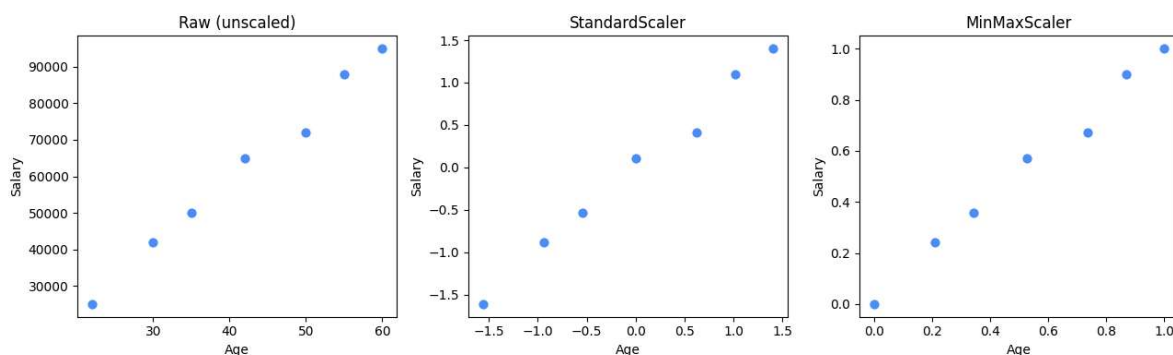
```
# Demo: age vs salary
data = pd.DataFrame({'age':[22,35,47,60,18,55,30,42,25,50],
'salary':[25000,50000,75000,95000,18000,88000,42000,65000,30000,72000],
'bought':[0,1,1,1,0,1,0,1,0,1]})
```

```
X = data[['age', 'salary']].values; y = data['bought'].values
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.25, random_s
```

```
# ■ Apply all three scalers
ss = StandardScaler()
mm = MinMaxScaler()
rb = RobustScaler()
X_std = ss.fit_transform(X_train) # fit on train, transform train
X_mm = mm.fit_transform(X_train)
X_rb = rb.fit_transform(X_train)
X_test_std = ss.transform(X_test) # only transform test, never fit
print(f'Original Age: {X_train[:,0].min():.0f} to {X_train[:,0].max():.0f}')
print(f'Standard Age: {X_std[:,0].min():.2f} to {X_std[:,0].max():.2f}')
print(f'MinMax Age: {X_mm[:,0].min():.2f} to {X_mm[:,0].max():.2f}')
```

```
Original Age: 22 to 60
Standard Age: -1.56 to 1.40
MinMax Age: 0.00 to 1.00
```

```
# ■ Visualise effect of scaling
fig, axes = plt.subplots(1,3,figsize=(13,4))
for ax,X_s,title in zip(axes,[X_train,X_std,X_mm],
    ['Raw (unscaled)', 'StandardScaler', 'MinMaxScaler']):
    ax.scatter(X_s[:,0],X_s[:,1],c='#4F8EF7',s=70,edgecolor='white')
    ax.set_title(title); ax.set_xlabel('Age'); ax.set_ylabel('Salary')
plt.tight_layout(); plt.show()
```



```
# ■ Effect on KNN decision boundary
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score
for name, Xtr, Xte in [('Unscaled', X_train, X_test),
    ('StandardScaler', X_std, X_test_std)]:
```

```
knn = KNeighborsClassifier(n_neighbors=3)
knn.fit(Xtr, y_train)
acc = accuracy_score(y_test, knn.predict(Xte))
print(f'{name:>15} KNN accuracy: {acc:.2f}')
```

```
Unscaled KNN accuracy: 1.00
StandardScaler KNN accuracy: 1.00
```

Lab — MNIST Handwritten Digit Recognition

What is MNIST?

MNIST contains 70,000 grayscale 28x28 pixel images of handwritten digits (0-9). Each image flattens to a vector of 784 pixel values (0-255). We treat each pixel as a feature — KNN classifies a new digit by finding the K most visually similar training images and taking a vote.

Why MNIST is a Good KNN Test

KNN works by geometric proximity — similar digits look visually similar, meaning their pixel vectors are close in 784-dimensional space. A handwritten '7' in the test set should sit near the '7' images in training. The challenge is that '1' and '7', or '4' and '9', can look similar — this shows up clearly in the confusion matrix.

Full Pipeline

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.datasets import fetch_openml
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
import seaborn as sns
```

```
import numpy as np
import matplotlib.pyplot as plt
# from sklearn.datasets import fetch_openml # Commenting out problematic imp
import tensorflow as tf # Import TensorFlow
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
import seaborn as sns

# ■ Load MNIST using tf.keras.datasets (more robust) ■
print('Loading MNIST dataset...')
(X_train_full, y_train_full), (X_test_full, y_test_full) = tf.keras.datasets
```

```
# Combine train and test for consistent processing as per original fetch_op
X = np.concatenate((X_train_full, X_test_full), axis=0)
y = np.concatenate((y_train_full, y_test_full), axis=0)

# Flatten the images from 28x28 to 784 features
X = X.reshape(X.shape[0], -1)

print(f'Full dataset: {X.shape}') # (70000, 784)
# Ensure y is integer type, as fetch_openml does
y = y.astype(int)
```

```
Loading MNIST dataset...
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-data/11490434/11490434 0s 0us/step
Full dataset: (70000, 784)
```

```
# Use 10,000 samples (full set takes ~10 min to predict)
X, y = X[:10000], y[:10000]
X_train, X_test, y_train, y_test = train_test_split(
X, y, test_size=0.2, random_state=42, stratify=y)
print(f'Train: {X_train.shape} Test: {X_test.shape}')
```

```
Train: (8000, 784) Test: (2000, 784)
```

```
# Scale pixels 0-255 to standardised values
scaler = StandardScaler()
Xtr_s = scaler.fit_transform(X_train.astype(float))
Xte_s = scaler.transform(X_test.astype(float))
```

```
# Train KNN (ball_tree is faster for high-dim data)
knn = KNeighborsClassifier(n_neighbors=5, metric='euclidean',
algorithm='ball_tree', n_jobs=-1)
knn.fit(Xtr_s, y_train)
```

▼ **KNeighborsClassifier** ⓘ ?

```
KNeighborsClassifier(algorithm='ball_tree', metric='euclidean', n_jobs=-1)
```

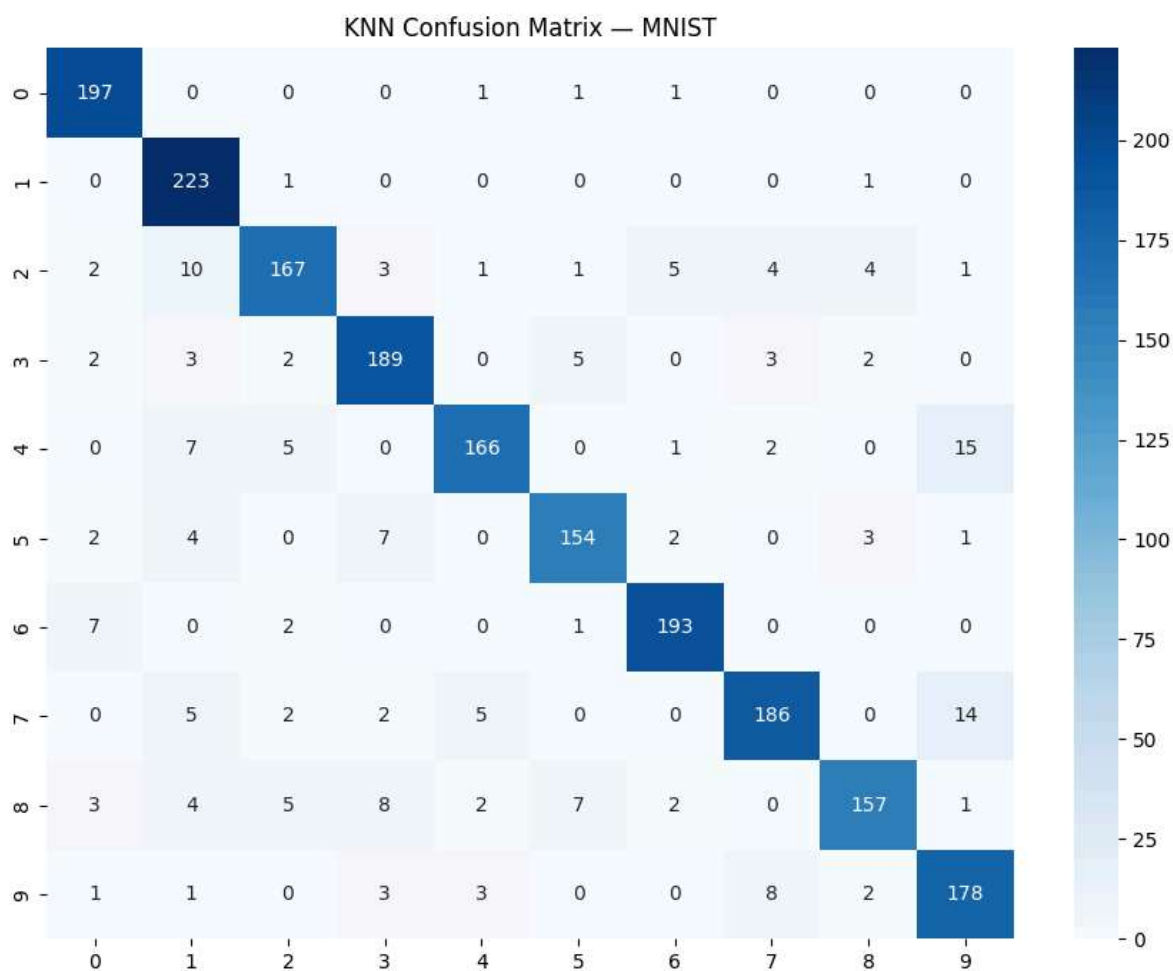
```
# Evaluate
y_pred = knn.predict(Xte_s)
print(f'Test Accuracy: {accuracy_score(y_test, y_pred)*100:.2f}%')
print(classification_report(y_test, y_pred))
```

```
Test Accuracy: 90.50%
              precision    recall  f1-score   support

0               0.92        0.98        0.95         200
1               0.87        0.99        0.93         225
2               0.91        0.84        0.87         198
3               0.89        0.92        0.90         206
4               0.93        0.85        0.89         196
5               0.91        0.89        0.90         173
6               0.95        0.95        0.95         203
```

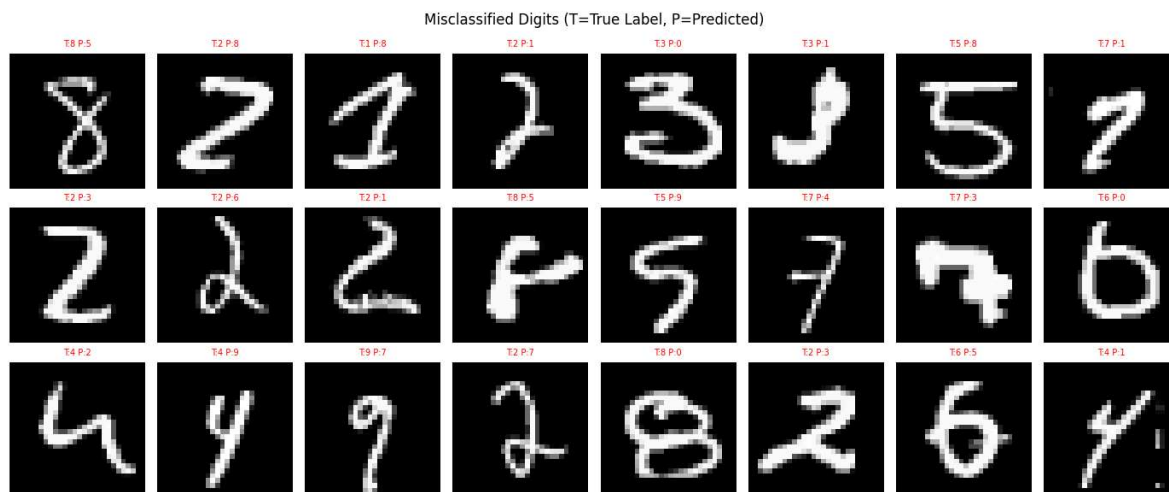
	7	0.92	0.87	0.89	214
	8	0.93	0.83	0.88	189
	9	0.85	0.91	0.88	196
accuracy				0.91	2000
macro avg	0.91	0.90	0.90	2000	
weighted avg	0.91	0.91	0.90	2000	

```
# ■ Confusion matrix
cm = confusion_matrix(y_test, y_pred)
plt.figure(figsize=(9,7))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',
xticklabels=range(10), yticklabels=range(10))
plt.title('KNN Confusion Matrix - MNIST'); plt.tight_layout(); plt.show()
```



```
# ■ Show misclassified images
wrong = np.where(y_pred != y_test)[0]
```

```
fig, axes = plt.subplots(3,8,figsize=(14,6))
for i,ax in enumerate(axes.flat):
    if i < len(wrong):
        ax.imshow(X_test[wrong[i]].reshape(28,28), cmap='gray')
        ax.set_title(f'T:{y_test[wrong[i]]} P:{y_pred[wrong[i]]}',
                    color='red', fontsize=7)
        ax.axis('off')
plt.suptitle('Misclassified Digits (T=True Label, P=Predicted)')
plt.tight_layout(); plt.show()
```



```
# ■ Compare K values
for k in [1,3,5,7,9]:
```